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Bachelor of Liberal Arts & Sciences

SSCECON307 ECONOMETRICS (FALL 2024)

How government spending effects inflation

(During economic crisis of 2008)

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Contents

1 Introduction **2**

1.1 Research question 2

1.2 Literature review 2

2 Data **4**

2.1 Datasets 4

2.2 Cleaning process 6

2.3 Final dataset 6

3 Methodology **8**

3.1 Equation 8

3.2 Hypothesis 9

3.3 Assumptions 9

3.3.1 Linearity 9

3.3.2 Multicollinearity 10

3.3.3 Heteroskedasticity: 11

3.3.4 Omitted variable bias 13

4 Results **14**

4.1 Model 1 14

4.2 Model 2 15

5 Limitations **16**

6 Conclusion **16**

7 References **17**

1 Introduction

1.1 Research question

During periods of economic crises, governments often play a pivotal role in steering the economy back to safety and prosperity. This significant involvement is due to governments possessing substantial resources compared to individual citizens and companies. When utilized effectively, these resources can aid in recovering the economy from losses.

However, one of the most critical challenges for governments during crises is to allocate spending in ways that are most beneficial for the economy. The magnitude and direction of government expenditure can have profound positive or negative effects on various sectors and the economy as a whole. For citizens, economic parameters such as inflation directly impact their daily consumption and quality of life.

This research paper aims to analyze the effect of government spending on inflation during periods of economic crises, specifically focusing on the 2008 financial crash and the subsequent two years.

Understanding how government actions influence inflation is crucial for government agencies. It enables them to develop effective strategies and have a ready playbook for similar future situations, ensuring that fiscal interventions do not inadvertently exacerbate economic challenges.

1.2 Literature review

Overall, the literature suggests that government spending does have a positive effect on inflation; however, during economic crises, this may not be entirely the case.
John Maynard Keynes posited that during economic downturns, active government intervention through increased spending is essential to boost aggregate demand and stimulate growth (Keynes, 1936).

Keynesian economists argue that in times of recession, private sector demand declines, and government spending can fill the gap to prevent further economic contraction. However, critics caution that excessive spending may lead to inflation if it surpasses the economy's productive capacity (Blinder & Zandi, 2010).

Monetarist theorists, led by Milton Friedman, contend that inflation is primarily a monetary phenomenon resulting from excessive growth in the money supply (Friedman, 1963). They argue that financing government deficits by printing money can devalue currency and lead to inflation. During the 2008 crisis, concerns were raised that large-scale fiscal stimulus and quantitative easing could ignite inflationary pressures (Taylor, 2009).

(Blinder & Zandi, 2010) analyzed the effects of the U.S. government's fiscal response to the 2008 crisis. Their study concluded that the stimulus measures prevented a deeper recession and did not result in significant inflationary pressures. The output gap—the difference between actual and potential economic output—remained wide, suggesting that the economy could absorb increased demand without accelerating inflation.

Similarly, (Christiano, Eichenbaum and Rebelo, 2011) found that fiscal multipliers are larger when nominal interest rates are at the zero lower bound, as they were during the crisis. Their research indicated that government spending had a positive effect on output with minimal impact on inflation.

International analyses reflect comparable findings. The International Monetary Fund (IMF) reported that in advanced economies, inflation remained subdued despite substantial fiscal expansions (IMF, 2012). Factors such as high unemployment and underutilized resources contributed to keeping inflation low, supporting the view that government spending did not pose immediate inflationary risks during the crisis.

2 Data

The research goal is to understand the effect of government spending on inflation. The following are the initial datasets, used for creating a final dataset with joined initial ones.

2.1 Datasets

Dependent Variable

Inflation is measured using the (World Bank Group, 2024) Deflator Index Growth Rate, which captures the annual percentage change in the price level of all domestically produced goods and services.

Independent Variable

Government spending is calculated as a percentage of GDP. (Our World in Data, 2024).

Control Variables

The control variables included in the analysis are grounded in literature and reflect factors that might influence the inflation.

1. **GDP Growth Rate (%)**

   A robust economy can influence inflation dynamics by boosting demand levels. According to (Blanchard , 2016), higher GDP growth is associated with increased inflation due to heightened economic activity and demand pressures. (World Bank Group, 2024)

2. **Interest Rates (%)**

   Interest rates represent the average central bank rates, which are pivotal in shaping borrowing costs and consumer spending. (Bernanke and Gertler, 1995) emphasize the role of central bank policies in maintaining inflation near target levels, making interest rates a critical variable for this analysis. (World Bank Group, 2024)

3. **Unemployment Rate (%)**

   The relationship between unemployment and inflation, known as the Phillips Curve, suggests that lower unemployment rates may lead to higher inflation. (Blanchard,

2016) highlights how tight labor markets can increase wages, contributing to inflation. (World Bank Group, 2024)

4. **Government Debt (% of GDP)**

   Higher levels of government debt can influence inflation, particularly in emerging markets. (Reinhart and Rogoff, 2010) find that in well-developed countries, there is no significant link between debt and inflation, while in emerging markets, debt levels can exacerbate inflationary pressures. (World Bank Group, 2024)

5. **Money Supply Growth (%)**

   The strong correlation between money supply growth and inflation has been well-established in the literature. (McCandless and Weber, 1995) demonstrate that monetary expansion significantly impacts inflation levels. (World Bank Group, 2024)

6. **Dummy Variable for Economic Development**

   A binary variable is included to capture structural differences in inflation responses during crises. (Taylor, 2009) argues that during the 2008 financial crisis, developed economies generally experienced disinflation or deflation due to collapsing demand and aggressive monetary easing, while developing economies faced inflation from currency depreciation and rising import costs. (IMF, 2008).

2.2 Cleaning process

The initial datasets required significant cleaning and formatting to ensure they were suitable for analysis. The final dataset was created by merging multiple datasets based on country and year of observation. It includes 26 countries, each with two years of observations (2008 and 2009) for the variables discussed earlier. The number of observations is limited because the initial datasets contained missing values for many countries during the specified time frame.

Additionally, Azerbaijan was excluded from the dataset due to extreme outlier values in its inflation data: nearly 28% inflation in 2008 followed by -19% in 2009. This dramatic fluctuation was an outlier from the patterns observed in other countries and could effect the analysis.

2.3 Final dataset

The final dataset consists of 25 observations from the following countries: Australia, Belarus, Bulgaria, Colombia, Georgia, Hungary, Iceland, Indonesia, Jamaica, Japan, Jordan, Malaysia, Mauritius, Moldova, New Zealand, Peru, Philippines, Romania, Singapore, Switzerland, Thailand, Ukraine, the United Kingdom, the United States, and Uruguay. Even though the sample size is limited and some economies differ the sample covers most of the world and can be generalized to the whole world.

Table 1: Summary Statistics for 2008

Variable	**Mean**	**SD**	**Min**	**Max**
Gov Spend (%)	26.9277	9.9316	13.5816	50.8153
Rate (%)	2.5838	7.0854	-11.8077	22.3239
Inflation (%)	8.8562	7.3845	-1.3863	29.0750
Debt (% of GDP)	51.6467	38.6263	10.6566	138.5743
Unemployment (%)	5.8280	3.7461	0.8000	17.8700
GDP Growth (%)	3.8132	3.4055	-1.2243	10.2000
Money Supply (M2, % GDP)	78.9217	45.0635	23.1807	198.3870
Developed (Dummy)	0.36	0.4899	0	1

Observations: 25

Table 2: Summary Statistics for 2009

Variable	**Mean**	**SD**	**Min**	**Max**
Gov Spend (%)	28.1276	9.5925	14.3040	44.9999
Rate (%)	7.0445	5.5624	-1.2231	20.4389
Inflation (%)	3.1677	4.1718	-5.9913	12.5818
Debt (% of GDP)	58.0079	42.9423	18.5575	156.9845
Unemployment (%)	7.1788	3.8764	1.4900	20.7100
GDP Growth (%)	-1.8666	4.5942	-15.1365	5.0237
Money Supply (M2, % GDP)	83.1765	49.4021	26.4851	216.0249
Developed (Dummy)	0.36	0.4899	0	1

Observations: 25

It is important to note that each observation is a different country and across two years same group of countries is analyzed.

From (Tables 1, 2), we observe that most of the means and standard deviations remain relatively consistent across the two years, except for inflation and GDP growth rates. The mean inflation decreases significantly from 2008 to 2009, reflecting notable changes in economic conditions. Additionally, the high standard deviation relative to the mean of inflation, indicates substantial variability in the data. This large variance suggests that it is important to separately analyze developed and developing countries in the regression to account for potential differences in their economic dynamics.

3 Methodology

3.1 Equation

The first model excludes the dummy variable. This way world economy as a whole can be analyzed. The second model includes the dummy variable, so the difference between developing and developed countries can be clearly seen.

The following variable names are used:

1. $inflation$: Inflation, measured by the World Bank GDP deflator index growth rate. (measured in %, dependent variable)
2. $gov_spend_percentage$: Government spending as a percentage of GDP (measured in %, independent variable). gdp_growth: GDP growth rate (measured in %, control variable).
3. $rate$: Average central bank interest rates (measured in %, control variable).
4. $percentage_unemploy$: Unemployment rate (measured in %, control variable).
5. $debt_percentage$: Government debt as a percentage of GDP (measured in %, control variable).
6. $M2$: Money supply: Broad Money as a % of GDP.

7. *Developed*: Dummy variable (1 for developed countries, 0 for developing countries)

The first model estimated using Ordinary Least Squares (OLS) is specified as follows:

$$\begin{aligned} inflation = \beta_0 + \beta_1 gov_spend_percentage + \\ + \beta_2 gdp_growth + \beta_3 rate + \beta_4 percentage_unemploy + \\ + \beta_5 debt_percentage + \beta_6 M2 \end{aligned} \tag{1}$$

The second model estimated using Ordinary Least Squares (OLS) is specified as follows:

$$\begin{aligned} inflation = \beta_0 + \beta_1 gov_spend_percentage + \\ + \beta_2 gdp_growth + \beta_3 rate + \beta_4 percentage_unemploy + \\ + \beta_5 debt_percentage + \beta_6 M2 + \beta_7 Developed \end{aligned} \tag{2}$$

3.2 Hypothesis

The hypotheses being tested in this research are the following:

$H_0 : \beta_1 = 0$ (Null Hypothesis: Government spending has no effect on inflation)

$H_1 : \beta_1 \neq 0$ (Alternative Hypothesis: Government spending influences inflation)

3.3 Assumptions

The analysis was conducted using a dataset of 23 countries for years 2008 and 2009, focusing on how government spending influences inflation.

For an OLS to be the optimal fitted model, several assumptions must hold true. Those are the following:

3.3.1 Linearity

To check whether the relationship is linear the predicted values of inflation were plotted against residuals. From the graphs (Figures 1,3,2,4) it can be seen that there are no patterns, moreover there are "clouds" of point around zero.

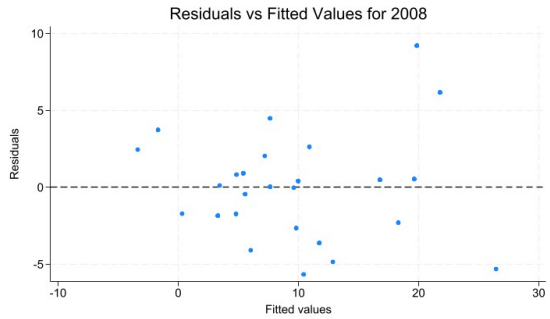


Figure 1: Graph Linearity 2008, Model 1

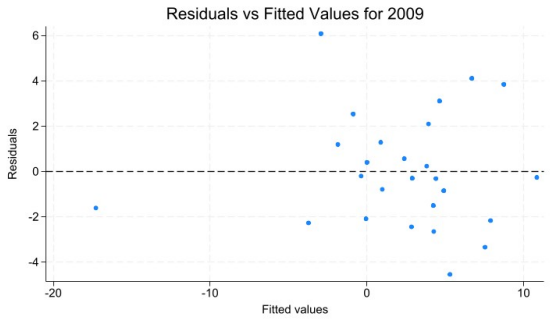


Figure 2: Graph Linearity 2009, Model 1

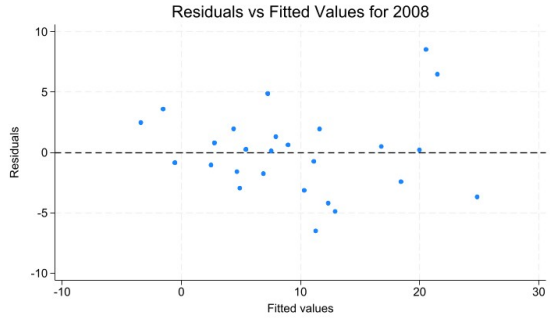


Figure 3: Graph Linearity 2008, Model 2

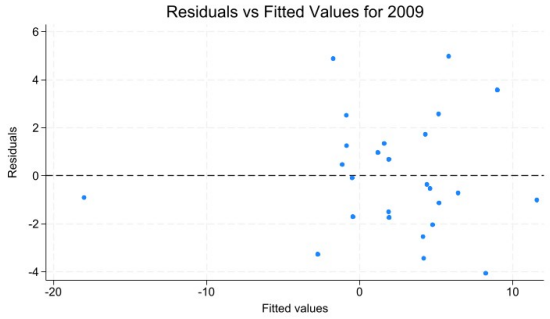


Figure 4: Graph Linearity 2009, Model 2

3.3.2 Multicollinearity

The Variance Inflation Factor (VIF) analysis (Tables 3, 4) reveals no significant multicollinearity issues, as all VIF values are below 2.75 with the mean VIF = 1.59 for year 2008 (model 1), VIF = 1.64 for year 2009 (model 1), VIF = 1.71 for year 2008 (model 2) and VIF = 1.69 for year 2009 (model 2). Thus it can be concluded that in both models there is multicollinearity.

Variable	VIF (2008)	1/VIF (2008)	VIF (2009)	1/VIF (2009)
M2	2.07	0.4820	1.86	0.5367
debt_percentage	1.98	0.5049	1.67	0.5985
gdp_growth	1.97	0.5072	1.79	0.5600
gov_spend_percentage	1.20	0.8355	1.89	0.5295
percentage_unemploy	1.15	0.8701	1.21	0.8292
rate	1.14	0.8790	1.40	0.7158
Mean VIF	**1.59**		**1.64**	

Table 3: Variance Inflation Factors (VIF) for the first model for 2008 and 2009

Variable	VIF (2008)	1/VIF (2008)	VIF (2009)	1/VIF (2009)
M2	2.37	0.4224	2.10	0.4764
gdp_growth	2.04	0.4894	1.81	0.5532
debt_percentage	1.98	0.5049	1.68	0.5943
developed	1.83	0.5467	1.59	0.6309
percentage_unemploy	1.34	0.7484	1.22	0.8209
gov_spend_percentage	1.25	0.8023	1.92	0.5199
rate	1.14	0.8786	1.52	0.6583
Mean VIF	**1.71**		**1.69**	

Table 4: Variance Inflation Factors (VIF) for Model 2 for 2008 and 2009

3.3.3 Heteroskedasticity:

To check for heteroskedasticity Breusch-Pagan Test was runned (Tables 5,6). For year 2008 (for both models) there is heteroskedasticity which is addressed below using robust regression. The reason for heteroskedasticity can be the structural difference between the economies in

a crisis year becomes even bigger. For year 2009 (for both models) the Test indicates that there is no heteroskedasticity and there is no need for robust regression.

Test Statistic	**2008**	**2009**
Variable	Fitted values of infl_percentage	Fitted values of infl_percentage
Assumption	Normal error terms	Normal error terms
chi2(1)	6.53	0.05
Prob > chi2	0.0106	0.8208
H0: Constant variance	**Rejected**	**Not Rejected**

Table 5: Breusch–Pagan/Cook–Weisberg Test Model 1 for Heteroskedasticity (2008 and 2009)

Test Statistic	**2008**	**2009**
Variable	Fitted values of infl_percentage	Fitted values of infl_percentage
Assumption	Normal error terms	Normal error terms
chi2(1)	5.39	0.49
Prob > chi2	0.0203	0.4839
H0: Constant variance	**Rejected**	**Not Rejected**

Table 6: Breusch–Pagan/Cook–Weisberg Test Model 2 for Heteroskedasticity (2008 and 2009)

Robust regression for 2008 (model 1 and model 2) (Table 7) shows that the government spending does not have statistically significant effect on inflation, while the m2 and interest rates are the ones that have statistically significant effects on inflation. These findings align with the results obtained in the OLS regression.

Variable	**Model 1 (2008)**	**Model 2 (2008)**
gov_spend_percentage	0.145 (0.093)	0.162* (0.106)
rate	-0.755** (0.152)	-0.752** (0.143)
debt_percentage	0.014 (0.028)	0.014 (0.028)
percentage_unemploy	0.067 (0.151)	-0.021 (0.141)
gdp_growth	0.496 (0.372)	0.441 (0.398)
m2	-0.077** (0.020)	-0.068** (0.019)
developed	—	-2.070 (2.144)
Constant	10.198* (4.247)	10.527** (4.319)
R-squared	0.8121	0.8204
Observations	25	25

Table 7: Robust Linear Regression Results for Model 1 and Model 2 for year 2008
Notes: * p < 0.10, ** p < 0.05, *** p < 0.01

3.3.4 Omitted variable bias

Based on the analyses conducted, there is no evidence to suggest the presence of omitted variable bias in either model. The consistency between the robust regression and OLS results for the year 2008, coupled with the fulfillment of key assumptions such as linearity, multicollinearity, and heteroskedasticity (the latter being addressed for both models in 2009), underscores the reliability of the findings. These results indicate that the models are well-specified and capable of adequately capturing the relationship between government spending and inflation while accounting for other relevant economic factors. Furthermore, the choice of control variables aligns with economic literature, further reinforcing the validity of the models and the robustness of the results.

4 Results

4.1 Model 1

The results of Model 1 (Table 8) indicate that government spending has no statistically significant effect on inflation in either 2008 or 2009. The two-tailed p-values for the gov_spend_percentage variable are well above the 0.10 significance level, suggesting that changes in government spending did not significantly influence inflation during these years.

However, the analysis reveals that the interest rate (rate) has a significant negative effect on inflation in both years, aligning with established economic theories. Higher interest rates tend to reduce inflation by decreasing consumer and business spending, thereby slowing down economic activity. This finding corroborates previous research and highlights the effectiveness of monetary policy in controlling inflation.

Similarly, the money supply variable (m2) shows a significant negative relationship with inflation. While traditional economic theory often suggests that an increase in money supply leads to higher inflation, the negative coefficient in this context may reflect the unique economic conditions during the 2008–2009 financial crisis. It's possible that increases in the money supply did not translate into higher inflation due to factors like decreased velocity of money or liquidity traps, where additional money fails to stimulate economic activity.

Additionally, the model's constant term is relatively high—approximately 10%—which corresponds with the elevated inflation levels during the crisis years. This aligns with global economic conditions at the time, as many countries experienced higher inflation rates due to economic instability and uncertainty.

Other variables such as debt_percentage, percentage_unemploy, and gdp_growth do not exhibit consistent significant effects on inflation across both years. Notably, gdp_growth has a positive coefficient in 2008 and a negative coefficient in 2009, with statistical significance in 2009. This shift may indicate changing dynamics in the relationship between economic growth and inflation during the crisis period, possibly due to differing fiscal and monetary

responses by governments.

The model demonstrates strong explanatory power, with R-squared values of 0.8121 for 2008 and 0.8213 for 2009, indicating that over 80% of the variability in inflation is explained by the model in both years.

Variable	**2008**	**2009**
gov_spend_percentage	0.145 (0.090)	0.020 (0.084)
rate	-0.755*** (0.121)	-0.479*** (0.070)
debt_percentage	0.014 (0.029)	0.033* (0.017)
percentage_unemploy	0.067 (0.237)	-0.061 (0.166)
gdp_growth	0.496 (0.316)	-0.327** (0.154)
m2	-0.077** (0.026)	-0.077*** (0.016)
Constant	10.198** (4.679)	10.218*** (2.934)
R-squared	0.8121	0.8213
Adjusted R-squared	0.7528	0.7648
Observations	25	25

Table 8: Regression Results for 2008 and 2009

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4.2 Model 2

The results of Model 2 (Table 9) exhibit similarities to those of Model 1, with some notable differences and slight improvements in the model's explanatory power. Specifically, Model 2 includes an additional variable, developed, which indicates whether a country is classified as a developed economy. This inclusion results in marginally higher R-squared and adjusted R-squared values compared to Model 1, suggesting that Model 2 explains a bit more of the variation in inflation.

Variable	**2008**	**2009**
gov_spend_percentage	0.162* (0.092)	0.035 (0.083)
rate	-0.752*** (0.121)	-0.506*** (0.072)
debt_percentage	0.014 (0.029)	0.035* (0.017)
percentage_unemploy	-0.021 (0.256)	-0.083 (0.164)
gdp_growth	0.441 (0.323)	-0.305* (0.152)
m2	-0.068** (0.027)	-0.069*** (0.016)
developed	-2.070 (2.269)	-1.957 (1.469)
Constant	10.527** (4.714)	10.200*** (2.876)
R-squared	0.8204	0.8373
Adjusted R-squared	0.7506	0.7741
Observations	25	25

Table 9: Linear Regression Results for 2008 and 2009

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5 Limitations

The main limitation of this research is the lack of a big sample size. Also, structural differences in economies can potentially influence the findings.

6 Conclusion

The lack of a significant effect of government spending on inflation could be due to the immediate impact of fiscal policies during the crisis not being reflected in inflation rates. Government spending may have been directed toward stabilizing financial systems and supporting employment, which might not have directly influenced price levels in the short term.

However, other results fully align (Taylor, J. B, 2009) and others, describing the reasons and consequences of the 2008 fall.

Future research can look into the effects of government spending on unemployment and other economic parameters, which than after a lag can influence the inflation in the long run.

7 References

Auerbach, A. J., & Gorodnichenko, Y. (2012). Measuring the Output Responses to Fiscal Policy. American Economic Journal: Economic Policy, 4(2), 1–27.

Blinder, A. S., & Zandi, M. (2010). How the Great Recession Was Brought to an End.

Christiano, L., Eichenbaum, M., & Rebelo, S. (2011). When Is the Government Spending Multiplier Large? Journal of Political Economy, 119(1), 78–121.

Coibion, O., & Gorodnichenko, Y. (2015). Is the Phillips Curve Alive and Well After All? Inflation Expectations and the Missing Disinflation. American Economic Journal: Macroeconomics, 7(1), 197–232.

Federal Reserve. (2014). Monetary Policy Report.

Friedman, M. (1963). Inflation: Causes and Consequences. Asia Publishing House.

International Monetary Fund. (2012). World Economic Outlook: Growth Resuming, Dangers Remain.

Keynes, J. M. (1936). The General Theory of Employment, Interest and Money.

Taylor, J. B. (2009). The Financial Crisis and the Policy Responses: An Empirical Analysis of What Went Wrong. NBER Working Paper No. 14631.

World Bank Group, 2024. Inflation consumer prices (annual%). Link

Our World in Data, 2024. Government spending as a percentage of GDP. Link

World Bank Group, 2024. GDP growth (annual %). Link

World Bank Group, 2024. Real interest rate (%). Link

World Bank Group, 2024. Unemployment, total (% of total labor force). Link

World Bank Group, 2024. Central government debt, total (% of GDP). Link

World Bank Group, 2024. Money supply growth (annual %). Link

International Monetary Fund, 2008. World Economic Outlook Database. Link